
Welcome to the
**NASA Breakthrough Propulsion Physics
Program**
Public Information Site



(Last update: 1999-Sept.-1)

In 1996, NASA established the Breakthrough Propulsion Physics program to seek the ultimate breakthroughs in space transportation: (1) propulsion that requires no propellant mass, (2) propulsion that attains the maximum transit speeds physically possible, and (3) breakthrough methods of energy production to power such devices. Topics of interest include experiments and theories regarding the coupling of gravity and electromagnetism, the quantum vacuum, hyperfast travel, and superluminal quantum effects. Because the propulsion goals are presumably far from fruition, a special emphasis is to identify affordable, near-term, and credible research that could make measurable progress toward these propulsion goals.

This web site describes the methods and activities of the Breakthrough Propulsion Physics program. For an introduction about the challenges of interstellar travel and some of the emerging concepts, visit our "[Warp Drive- When?](#)" site.

The Breakthrough Propulsion Physics program is managed by [Marc G. Millis](#) of the [Glenn Research Center \(GRC\)](#), and is sponsored jointly by the [Advanced Space Transportation Program](#), managed by [NASA Marshall Space Flight Center](#), Huntsville, AL, and the [Advanced Concepts Program](#) of the [NASA Office of Space Science](#), Washington, DC.

CONTENTS

- [What's **NEW**](#)
- [Program Descriptions](#)
- [Publications](#)
- [Research](#)

- [Conferences / Workshops](#)
- [Links to Other Government Activities](#)
- [Questions and Answers](#)

READ THESE FIRST

The quickest way to get an overall sense of what this program is about is to read the following two items:

- [NASA/TM-1998-208400](#) (only about 10 pages long including reference list)
 - [Recent research proposals selections](#)
-
-

WHAT's NEW (1999-Aug-20):

- [1999, 08, 19] [News Release: "Research Selections"](#)
- [1999, 08, 19] [Short descriptions of selected proposals](#)
- [1999, 02] Call closes for research proposals ([NRA-99-LeRC-1](#))
- [1999, 01] Workshop proceedings published (NASA CP-1999-208694)
- [1999, 01] BPP program gets top score from Independent Review Panel
- [1998, 11] Call opens for research proposals ([NRA-99-LeRC-1](#))

PROGRAM DESCRIPTIONS

The following items describe stages of the program in reverse chronological order:

- [1999, 08] [Research proposals selected](#)
- [1998, 11] Soliciting research proposals via NASA Research Announcement ([NRA-99-LeRC-1](#))
- [1998, 10] [White Paper](#)
- [1998, 06] Description of program goals, methods, and emerging opportunities ([NASA/TM-1998-208400](#))
- [1997, 11] Determining if affordable research opportunities exist via workshop ([NASA TM-97-206241](#))
- [1996, 12] Setting goals, methods and identifying scientific foundations (NASA TM-107381)
- [1996] Program invitation and formulation

These items predate the official NASA program:

- [1994-1996] Grass-roots collaboration - [Interstellar Propulsion Society](#)
- [1994] "Practical Robotic Interstellar Flight: Are We Ready?", Conference, New York, NY
- [1990-1995] Lewis Research Center "Space Coupling Working Group"
- [1989-1993] Lewis Research Center "Vision-21" Group (NASA CP-10059, and NASA CP-10129)

Essays:

- [1997] [Why Now?](#)
- [1996/1997] [Enabling Interstellar Missions](#) - why interstellar travel is so difficult, and what is needed to overcome the challenges.

PUBLICATIONS

The following list shows some of the most recent publications related to the BPP program. In those cases where electronic copies are available, links are indicated. To get paper copies of US Government reports, contact the [National Technical Information Service \(NTIS\)](#) (There is a modest per-page fee associated with copies.)

- El-Genk, M. S., ed. (1999) "Space Technology and Applications International Forum 1999," American Institute of Physics, AIP Conference Proceedings 458. (The papers on BPP topics are in pages 875-1059.)
- Millis and Williamson, ed., (1999) "NASA Breakthrough Propulsion Physics Workshop Proceedings," NASA/CP-1999-208694, Proceedings of a conference held at and sponsored by NASA Lewis Research Center in Cleveland Ohio, August 12-14, 1997. (Jan. 99) (456 pg.).
[NOTE: a short workshop summary, NASA TM-97-206241, is available electronically.](#)
- [Millis \(1998\) "NASA Breakthrough Propulsion Physics Program", NASA/TM-1998-208400](#)
- Li, N., Noever, D., Robertson, T., Koczor, R., and Brantley, W. (1997) "Static Test for a Gravitational Force Coupled to Type II YBCO Superconductors," In Physica C, Vol. 281, pp. 260-267.
- [Millis \(1997\) "Breakthrough Propulsion Physics Workshop Preliminary Results," NASA TM-97-206241](#)
- [Millis \(1997\) "Challenge to Create the Space Drive," Journal of Propulsion and Power, Vol. 13, pp. 577-582](#)
- Millis (1996) "Breakthrough Propulsion Physics Research Program," NASA TM-107381
- Forward, R., (1996) "Mass Modification Experiment Definition Study," Rpt # PL-TR-96-3004, Phillips Lab, Edwards, CA.
- Millis, M. and Williamson, G.S., (1995) "Experimental Results of Hooper's Gravity-Electromagnetic Coupling Concept", NASA TM 106963.

RESEARCH

- [1999, 08, 19] [News Release: "Research Selections"](#)
- [1999, 08, 19] [Short descriptions of selected proposals](#)
- [1998] [Short reference list](#) (from NASA TM 97-206241)
- [1997] [Assessing the unsolved challenges of propulsion without propellant](#)

CONFERENCES / WORKSHOPS

The following conferences and workshops, listed in reverse chronological order, have contained at least one session related to the BPP program.

Pending:

- [2000, July, 17-19] [36th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit](#), Huntsville, AL ([CALL FOR PAPERS OPEN UNTIL 1999 NOV 8](#)).
- [2000, Jan. 30 - Feb. 3] [Space Technology Applications International Forum \(STAIF-2000\)](#), Albuquerque NM, specifically the "[Conference on Enabling Technologies and Required Scientific Developments for Interstellar Missions](#)"

Completed:

- [1999, June 20-24] 35th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit, Los Angeles, CA
- [1999, Jan. 31-Feb. 4] Space Technology Applications International Forum (STAIF-1999), Albuquerque NM.
- [1998, July 13-15] 34th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit, Cleveland, OH
- [1998, Oct. 6-8] BPP Lecture Series, [Ohio Aerospace Institute](#), Cleveland OH
- [1998, Feb. 9-12] Physics for the Third Millennium (Lecture Series), Marshall Space Flight Center, Huntsville AL
- [1998, Jan.] Space Technology Applications International Forum (STAIF-1998), Albuquerque NM.
- [1997 Aug.] [NASA Breakthrough Propulsion Physics Workshop](#), Cleveland OH
- [1997] BPP Brainstorming Session, Austin TX

LINKS TO OTHER GOVERNMENT ACTIVITIES

- [NASA Home Page](#)
- [NASA Glenn Research Center at Lewis Field](#)
 - [Programs and Projects at Glenn](#)
 - [Pioneering Ion Propulsion Research](#)
 - [Space Propulsion Research](#)
 - [Microgravity Research](#)
- [NASA Space Transportation Programs](#) (NASA MSFC)
- [NASA Office of Space Science](#) (NASA HQ)
- [Advanced Concepts Program](#) (NASA Office of Space Science)
- [Advanced Propulsion Concepts Notebook](#) (NASA JPL)
- [NASA Origins Program](#)
- [Gravity Probe B](#)
- [Laser Interferometer Gravitational-Wave Observatory](#)
- Interstellar Probe Missions:
 - [Missions under consideration by the NASA Office of Space Science](#)
 - [Technology options under study by NASA JPL](#)
- [NASA Institute for Advanced Concepts](#)

QUESTIONS AND ANSWERS

This section is temporarily unavailable

Author: Marc G. Millis

NOTE: Due to the overwhelming number of inquiries Mr. Millis receives, he regrets that he will not be able to respond to all inquiries. Please check first to see if your questions have already been addressed in the Q&A section before sending any correspondence to Mr. Millis.

If you have problems reading or browsing this web site email Marc.G.Millis@grc.nasa.gov
